JOHN MUNGER
AI Platform Engineer | Systems Brief
RAG Systems • LLM Orchestration • AI Runtime Infrastructure

Overview

I build AI platform systems that integrate LLMs into production workflows, focusing on retrieval grounding, orchestration, and runtime control, with an emphasis on making these systems reliable, observable, and cost-efficient.

AI Control Plane Runtime

Built a modular AI control plane runtime to manage interactions between application services and multiple LLM providers, acting as a centralized routing, policy, and evaluation layer.

Key Capabilities

- Model routing across providers based on task complexity, latency, and cost
- Centralized observability layer for controlled experimentation and monitoring
- Evaluation hooks to detect regressions and analyze response quality
- Extensible architecture supporting agent workflows and tool-calling systems

System Architecture

```
     +-----------------------------+
     |   Client / Application    |
     +--------------+--------------+
                    v
     +--------------+--------------+
     |     AI Control Plane      |
     +--------------+--------------+
                    v
     +--------------+--------------+
     |       Model Router        |
     |       Policy Layer        |
     |     Evaluation Hooks      |
     +--------------+--------------+
                    v
 +------------+--------------+------------+
 |   OpenAI   |  Anthropic   | Local LLM |
 +------------+------+--------+------------+
                    v
           +--------+---------+
           |    Response     |
           +------------------+
```

Enabled consistent experimentation and cost-aware optimization across models while maintaining a stable interface between application logic and AI systems.

Retrieval-Augmented Developer Assistant

Built a retrieval-grounded developer assistant to enable fast, accurate navigation of large documentation sets.

RAG Pipeline

```
  +-----------------------------------+
  |        User Query                 |
  +-----------------+-----------------+
                    v
+-------------------+-------------------+
|  Retriever (Vector Search)            |
+-------------------+-------------------+
                    v
+-------------------+-------------------+
|   Top-K Retrieved Chunks              |
+-------------------+-------------------+
                    v
+-------------------+-------------------+
|       Context Builder                 |
+-------------------+-------------------+
                    v
+-------------------+-------------------+
|   LLM (via Control Plane)             |
+-------------------+-------------------+
                    v
+-------------------+-------------------+
|     Grounded Response                 |
+-------------------+-------------------+
```

Architecture

- Document ingestion + embedding pipeline for structured knowledge indexing
- Vector search retrieval + context construction for grounded prompting
- Multi-model routing balancing response quality, latency, and cost

Impact

- Faster knowledge discovery across large documentation sets
- Reduced context switching and manual search effort
- Improved answer reliability through retrieval-grounded responses

Key Engineering Themes

Retrieval-first design • Cost-aware routing • Evaluation + observability • Platform-oriented AI architecture

Technical Stack

TypeScript • Node.js • Vector Search • Embeddings • RAG Pipelines • LLM APIs • Backend Services • Distributed Systems

Focus Areas

AI Platform Engineering • LLM Systems • Retrieval Systems • AI Runtime Infrastructure